

Positive-Noise Hilbert–Schmidt Analyticity and Local Shell Transport

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Abstract

Although the natural Hilbert–Schmidt family diverges at zero noise, it is analytic on every compact positive-noise interval. We use this to prove a local shell theorem: an exact isolated spectral cluster persists under one common contour with a uniform resolvent bound, and an exterior selector below the unit radius has a locally contractive quotient power. Thus the RH-269 package is automatic locally at positive noise. The theorem does not give a single rank-growing zero-noise selector.

1 Analytic family

For $0 < \sigma_- < \sigma_+ < \infty$, the folded Gaussian kernel and every σ derivative are continuous and uniformly bounded on $[0, 1]^2 \times [\sigma_-, \sigma_+]$. Its row normalizer is uniformly positive.

Theorem 1.1 (positive-noise $\mathcal{S}_2$ analyticity). *The map $\sigma \mapsto A_\sigma$ is real analytic from $(0, \infty)$ into $\mathcal{S}_2(L^2(0, 1))$. In particular, on every compact positive-noise interval, $\|A_\sigma - A_\tau\|_2 \leq L|\sigma - \tau|$ for some finite L .*

Proof. Fix $\sigma_0 > 0$ and complexify the noise parameter in a disk avoiding zero. The unnormalized folded kernel and its row normalizer are holomorphic there. At σ_0 the normalizer has a positive minimum over $x \in [0, 1]$; shrinking the disk keeps its complex modulus uniformly away from zero. The normalized kernel is therefore a bounded $L^2([0, 1]^2)$ -valued holomorphic map, by dominated difference quotients. Restriction to the real axis is real analytic. A finite cover of a compact positive-noise interval bounds the first derivative and gives the mean-value estimate. $\square$

2 Local shell activation

Fix $\sigma_0 > 0$ and a circle $\Gamma = \{|z| = r\}$, $0 < r < 1$, disjoint from $\text{spec}(A_{\sigma_0})$. Let E_{σ_0} be the range of the Riesz projection for the finite exterior spectrum $|z| > r$. Nearby, write E_σ for the transported Riesz range, Q_σ for the orthogonal projection onto $E_\sigma^\perp$, and $C_\sigma = Q_\sigma A_\sigma Q_\sigma|_{E_\sigma^\perp}$.

Theorem 2.1 (local exterior quotient). *For σ near σ_0 , the same circle is in the resolvent set, the exterior Riesz rank is constant, and*

$$\sup_{\sigma, z \in \Gamma} \|(z - A_\sigma)^{-1}\| < \infty.$$

The orthogonal quotient compressions C_σ vary in $\mathcal{S}_2$, have spectral radius below r , and possess a uniform contractive power on a smaller neighborhood. Hence the RH-269 tail theorem applies locally.

Proof. Let $M_0 = \sup_{z \in \Gamma} \|(z - A_{\sigma_0})^{-1}\|$. If $M_0\|A_\sigma - A_{\sigma_0}\| < 1$, the Neumann identity gives the common resolvent and bound $M_0/(1 - M_0\|A_\sigma - A_{\sigma_0}\|)$. Riesz ranks and orthogonal projections then vary continuously. Relative to $E_\sigma \oplus E_\sigma^\perp$, invariance of the Riesz range makes A_σ block upper triangular with C_σ as the interior diagonal block. Hence its spectrum is the interior spectral component and its radius is below $r < 1$. Gelfand's formula supplies a contractive power at σ_0 , stable on a smaller neighborhood by continuity. $\square$

3 Boundary

This is a local fixed-rank chart. It neither glues the rank-growing noisy cloud to zero nor proves the coefficient bridge. Gates A–E remain open.